

Lecture – 15

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In this lecture...

- i. Image Analysis
- ii. Feature Extraction
- iii. Types of Features
- iv. Detection of Discontinuities
- v. Segmentation
- vi. Discontinuities-based segmentation
 - i. Point Detection
 - ii. Line Detection
 - iii. Edge Detection

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Image Analysis

- Image Analysis refers to the process of extracting meaningful information from digital images.
- Major Steps in Image Analysis:
 - **Preprocessing** – Noise reduction, normalization, contrast enhancement.
 - **Segmentation** – Dividing the image into meaningful regions (e.g., foreground/background).
 - **Feature Extraction** – Describing parts of the image with quantitative descriptors.
 - **Object Recognition/Classification** – Assigning labels or categories.
 - **Post-processing** – Smoothing results, removing false detections, etc.

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Feature Extraction

- Feature extraction: the process of transforming image data into a set of characteristics (features) that effectively represent the image for a specific task like classification or detection.
- Features reduce the image's complexity while retaining essential information.
- General Process for feature extraction includes:
 - **Feature Detection:** Detect regions of interest (ROI) or keypoints.
 - **Feature Description:** Extract feature descriptors from those regions.
 - Store features as numerical vectors for further analysis.

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Local vs Global Features

- A local feature refers to a descriptor that is applicable to individual elements or regions within an image. These features are used to describe small parts or segmented components of the image independently.
- Examples:
 - Area of a bright region representing the amount of liquid in a single bottle on an assembly line.
 - Centroid, orientation, compactness, or eccentricity of a specific object in the image.
 - Corners, edges, or keypoints detected in small sub-regions.
- In object-level analysis or classification, local features are used to differentiate one object from another, such as detecting whether each bottle is full by analyzing the bright region inside it.
- A feature is considered local when it applies to a "member" of a set e.g., an object, region, or pixel within an image

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Local vs Global Features

- A global feature is a descriptor that applies to the entire image or a complete dataset of images. These features summarize the image content as a whole, rather than its individual parts.
- Examples:
 - Total area covered by bright pixels in the entire image (e.g., total liquid across all bottles).
 - Mean intensity, entropy, or Fourier descriptors computed over the full image.
 - Principal components or histogram-based descriptors summarizing the entire image's data.
- In scene classification or global similarity tasks (e.g., estimating total liquid passed in a day or classifying satellite images), global features give a single descriptor value per image.
- A feature is global when it applies to the entire set or image. This scope can shift depending on the application. For example, area can be local if describing one region or global if describing the entire image.

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Local vs Global Features

- **Discontinuities** refer to abrupt changes in intensity, and their detection is crucial for identifying **edges**, **lines**, and **corners** in images.
- These are the **primitives** of an image that represent boundaries, object shapes, and structural information.
- **Types of Discontinuities:**
 - **Point Discontinuities:** A sudden change at a single pixel (e.g., noise spike).
 - **Line Discontinuities:** A linear edge separating two regions.
 - **Edge Discontinuities:** Sharp changes in intensity often correspond to object boundaries.

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Image Segmentation

- Segmentation divides an image into its constituent parts or objects
- Level of subdivision depends on the problem being solved
- Segmentation stops when objects of interest in an application have been isolated
- Example: A disaster response system may be interested in identifying collapsed buildings in a satellite image after an earthquake.
 - **Segment the urban area from the satellite image** - Identify and isolate the regions where buildings and man-made structures are concentrated.
 - **Segment the contents of the urban area down to structures of varying sizes that correspond to buildings** - Analyze the segmented urban area to detect individual buildings, intact or damaged, based on size, shape, and texture.
 - **No need to segment natural landscapes or rural surroundings outside the urban area** - Focus computational resources only within urban zones to efficiently identify potential collapsed structures.

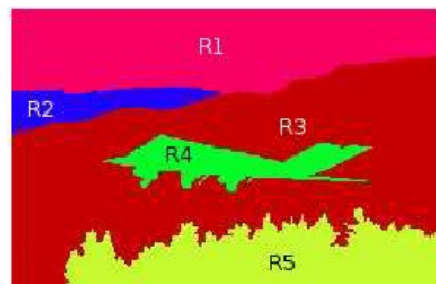
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Image Segmentation

- Segmentation is the first step in any automated computer vision application
- It attempts to partition the pixels of an image into groups that strongly correlated with the objects in an image.



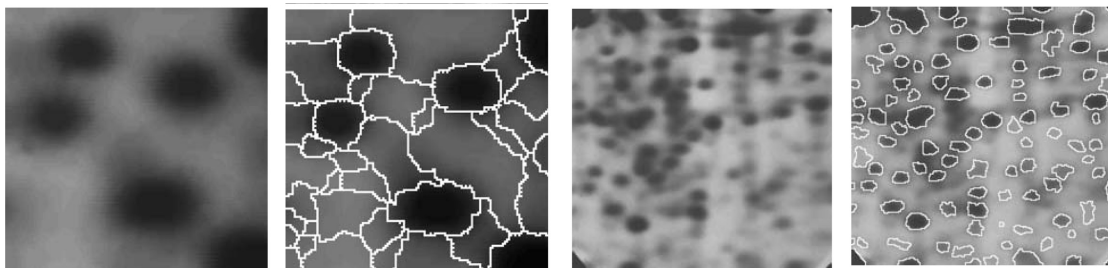
(a) Original Image



(b) Segmented Image

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Image Segmentation



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Image Segmentation

- Segmentation algorithm for monochrome images are based on one of two basic properties of gray-level values
 - Discontinuities
 - Similarities
- For Discontinuity, the approach is to partition an image based on abrupt changes in gray level
- The principal areas of interest are:
 - Detection of isolated points
 - Detection of lines and edges in an image
- Masks and Correlation (spatial filtering) are used to find discontinuities.

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Detection of Discontinuities – Point Detection

- A mask used for detecting isolated points (different from a constant background) would be

-1	-1	-1
-1	8	-1
-1	-1	-1

The response equation:

$$R = w_1z_1 + w_2z_2 + \dots + w_9z_9$$

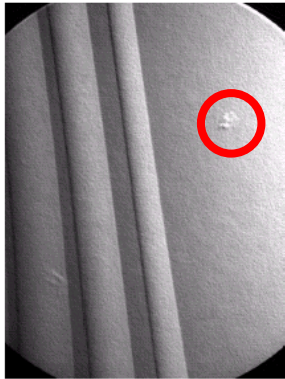
$$= \sum_{i=0}^9 w_i z_i$$

- An isolated point is detected if the response of the mask is greater than a predetermined threshold

$$|R| > T$$

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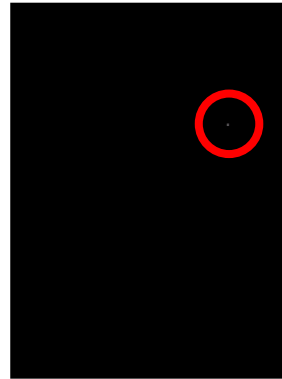
Detection of Discontinuities – Point Detection



X-ray image of a turbine blade



Result of point detection



Result of thresholding

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Detection of Discontinuities – Line Detection

- Given masks are used to extract lines that are one pixel thick and running in a particular direction.

-1	-1	-1
2	2	2
-1	-1	-1

Horizontal

2	-1	-1
-1	2	-1
-1	-1	2

+45°

-1	2	-1
-1	2	-1
-1	2	-1

Vertical

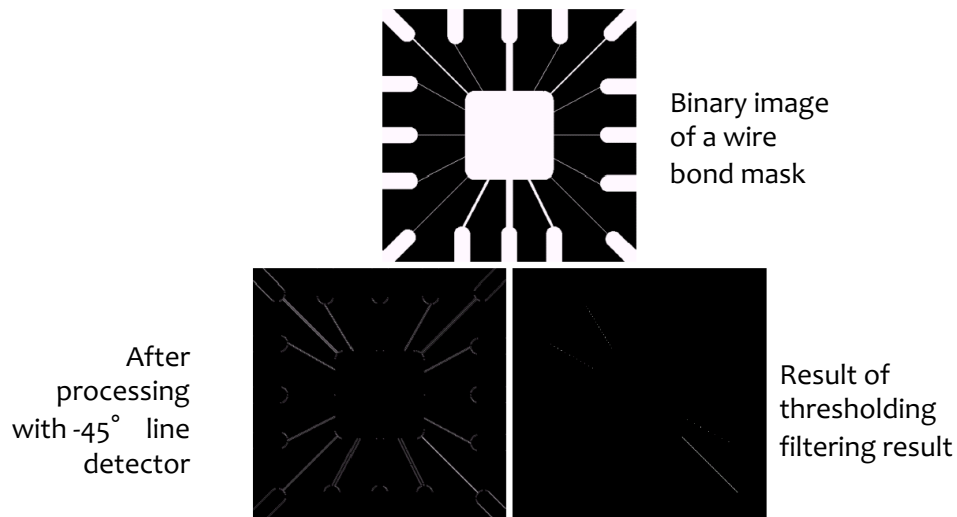
-1	-1	2
-1	2	-1
2	-1	-1

-45°

- With a constant background, the maximum response occurs when the line is “lined up” with the center of the mask. Noted that the preferred direction of each mask is weighted with a larger coefficient than other possible directions
- Let R_1 , R_2 , R_3 and R_4 are the response of the masks. And if at a certain point in the image, $|R_i| > |R_j|$ for all $j \neq i$ that point is said to be more likely associated with a line in the direction of mask i .

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Detection of Discontinuities – Line Detection



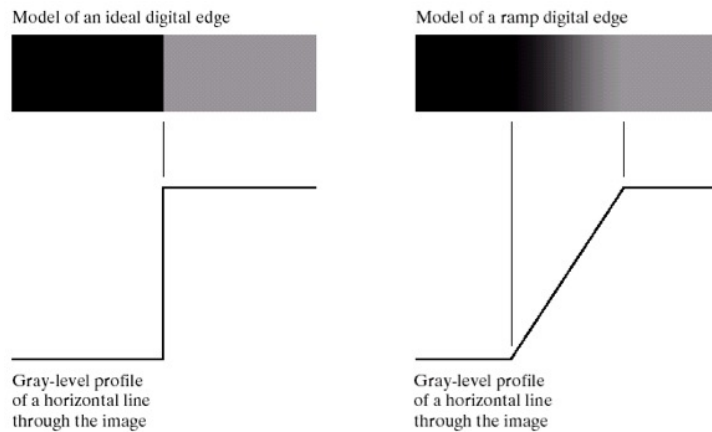
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Detection of Discontinuities – Edge Detection

- Edge detection is by far the most common approach for detecting discontinuities in gray levels
- Basic formulation and initial assumptions for edge detection are:
 - An edge is a boundary between two image regions with relatively distinct gray level properties.
 - Regions are sufficiently homogeneous so that the transition between the regions can be determined on the basis of gray-level discontinuities alone
 - If this is not valid, some other techniques will be used.
- The basic idea behind most edge detection techniques is the computation of a local derivative operator

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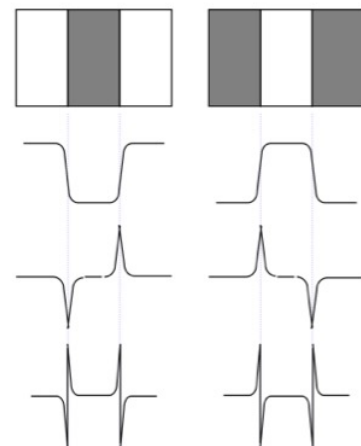
Detection of Discontinuities – Edge Detection



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Edge Detection – Derivative Operations

1. Image of a dark stripe on a light background (and vice versa)
2. A profile of the lines in the image modeled as a gradual rather than sharp transition – Edges in images tend to be slightly blurred as a result of sampling
3. The first derivative: The magnitude detects the presence of an edge
4. The second derivative: the sign tells the type of transition (light-to-dark or dark-to-light) and zero crossing at each edge



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Edge Detection – Derivative Operators

- The first derivative at any point in an image is computed using the magnitude of the gradient

$$\nabla f = \begin{bmatrix} G_x \\ G_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

- Where the magnitude is

$$\nabla f = \text{mag}(\nabla f) = [G_x^2 + G_y^2]^{1/2} = \left[\left(\frac{\partial f}{\partial x} \right)^2 + \left(\frac{\partial f}{\partial y} \right)^2 \right]^{1/2}$$

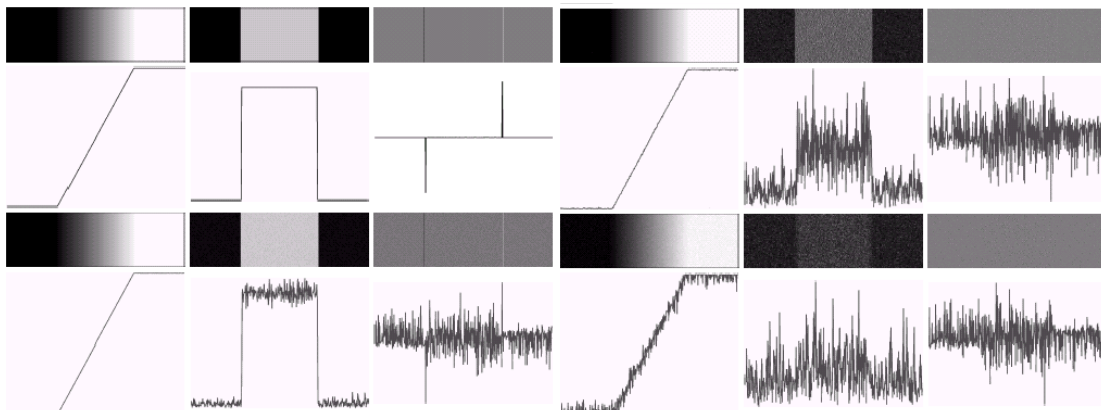
- The direction of gradient vector is the angle $\alpha(x, y)$ given by

$$\alpha(x, y) = \tan^{-1} \left(\frac{G_y}{G_x} \right)$$

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Edge Detection – Derivative Operators

Derivative based edge detectors are extremely sensitive to noise.



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Common Edge Detectors

- Given the 3*3 region of an image the following edge detection filter can be used

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

-1	-1	-1
0	0	0
1	1	1

-1	0	1
-1	0	1
-1	0	1

Prewitt

-1	0
0	1

0	-1
1	0

Roberts

-1	-2	-1
0	0	0
1	2	1

-1	0	1
-2	0	2
-1	0	1

Sobel

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Edge Detector Example



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Edge Detection Issues

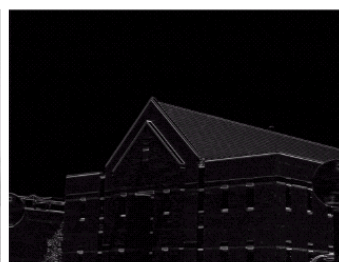
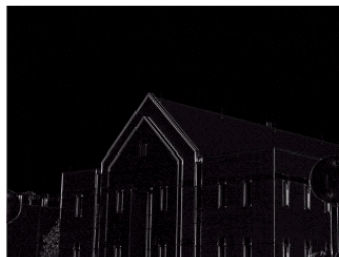
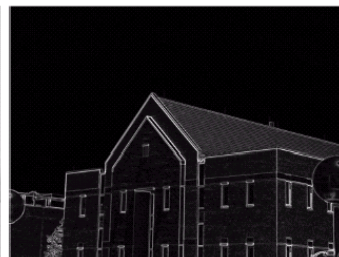
- When there is too much detail in the image, edge detection becomes unrealistic.
- One way to overcome this situation is to smooth images prior to edge detection.



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Edge Detector Example with Smoothing

Original Image

Horizontal
Gradient
ComponentVertical
Gradient
ComponentCombined
Edge Image

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Laplacian Edge Detection

- The 2nd – order derivative filter by Laplacian

0	-1	0	-1	-1	-1
-1	4	-1	-1	8	-1
0	-1	0	-1	-1	-1

- Laplacian filter is rarely used in edge detection because:
 - As a second derivative operator it is typically unacceptably sensitive to noise
 - The Laplacian produces double edges
 - Unable to detect direction
- Laplacian is more often use in secondary role of detector for establishing whether a pixel is on the light or dark side of an edge i.e. to find the location of edges using the zero crossing property.

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Laplacian Edge Detection

- Laplacian is combined with Gaussian low pass filter for edge detection.
- The basic idea is to convolve an image with the Laplacian of a 2-D Gaussian function of the form

$$G(x, y) = \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right)$$

σ = standard deviation

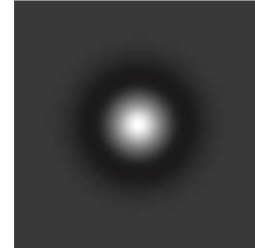
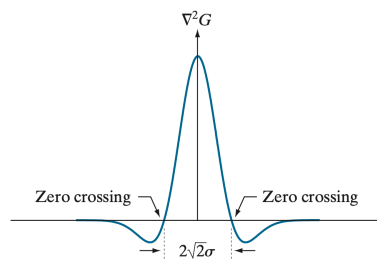
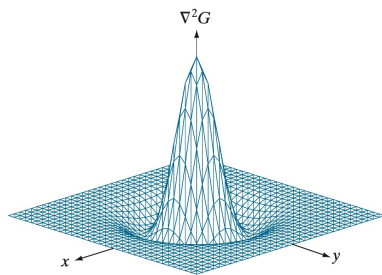
- If $r^2 = x^2 + y^2$, then the Laplacian is then

$$\nabla^2 G(x, y) = \left(\frac{x^2 + y^2 - 2\sigma^2}{\sigma^4}\right) e^{-\frac{x^2 + y^2}{2\sigma^2}}$$

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Laplacian Of Gaussian

- The Laplacian of Gaussian (or Mexican hat) filter uses the Gaussian for noise removal and the Laplacian for edge detection.



0	0	-1	0	0
0	-1	-2	-1	0
-1	-2	16	-2	-1
0	-1	-2	-1	0
0	0	-1	0	0

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Discussions and Classwork

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Lecture – 16

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In this lecture...

- i. Edge Linking
 - i. Local Processing
 - ii. Global Processing – Hough Transform
- ii. Thresholding
 - i. Global, Local, Dynamic Thresholding
 - ii. Adaptive Thresholding
 - iii. Optimal Thresholding – Otsu's Algorithm
 - iv. Using Edges to Improve Thresholding

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Edge Linking and Boundary Detection

- Ideally, edge detection techniques yield pixels lying only on the boundaries between regions
- In practice, such pixel set rarely characterizes a boundary completely because of
 - Noise
 - Breaks in the boundary due to non-uniform illuminations
 - Other effects that introduce spurious discontinuities
- Thus, edge detection algorithms are usually followed by linking and other boundary detection procedures designed to assemble edge pixels into meaningful boundaries.

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Edge Linking – Local Processing

- Basic idea is
 - analyze the characteristics of pixels in a small neighborhood (3*3, 5*5, etc) for every point (x,y) that has undergone edge detection
 - All points that are “similar” are linked, forming a boundary of pixels that share some common property
- Two principal properties for estimating similarity
 - The strength of the response of the gradient operator used to produce the edge pixels
 - The direction of the gradient

$$\alpha(x, y) = \tan^{-1} \left(\frac{G_y}{G_x} \right)$$

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Edge Linking – Local Processing

- An edge pixel at (x',y') in the neighborhood centered at (x,y) is similar in magnitude to the pixel at (x,y) if

$$|\nabla f(x,y) - \nabla f(x',y')| \leq T$$

where T is a predetermined threshold

- An edge pixel at (x',y') in the neighborhood centered at (x,y) is similar in angle to the pixel at (x,y) if

$$|\alpha(x,y) - \alpha(x',y')| \leq A$$

where A is a predetermined angle threshold

- A point in neighborhood of (x,y) is linked to (x,y) if both magnitude and angle criteria are satisfied.

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Edge Linking – Local Processing



- (a) Image of the rear of a vehicle.
- (b) Gradient magnitude image.
- (c) Horizontally connected edge pixels.
- (d) Vertically connected edge pixels.
- (e) The logical OR of (c) and (d).
- (f) Final result, using morphological thinning.

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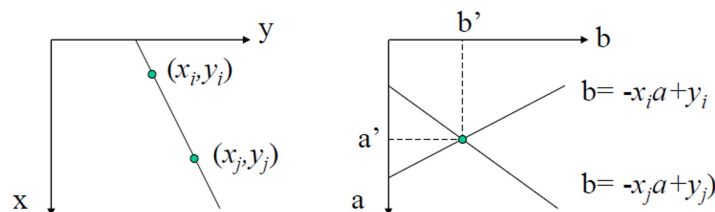
Edge Linking – Global Processing

- In this section, the global relationship between pixels are considered.
- Suppose that, for n points in an image, we want to find all subsets of these points that lie on straight lines
- Consider a point (x_i, y_i) and the equation for a straight line $y_i = ax_i + b$
- Infinitely many lines pass through the point (x_i, y_i) all satisfying the equation for varying values of a and b
- However, writing the equation as $b = -x_i a + y_i$ and considering the ab plane (also called the parameter space) yield the equation of single line for a fixed point (x_i, y_i)

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Global Processing – Hough Transform

- A second point (x_j, y_j) also has a line in the parameter space associated with it
- This line intersects the line associated with (x_i, y_i) at (a', b')
 - a' is the slope and b' is the intercept of the line containing both (x_i, y_i) and (x_j, y_j) in xy plane
- In fact, all points that lie on this line have corresponding lines in the parameter space that intersect at (a', b')



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Hough Transform – Accumulator Cells

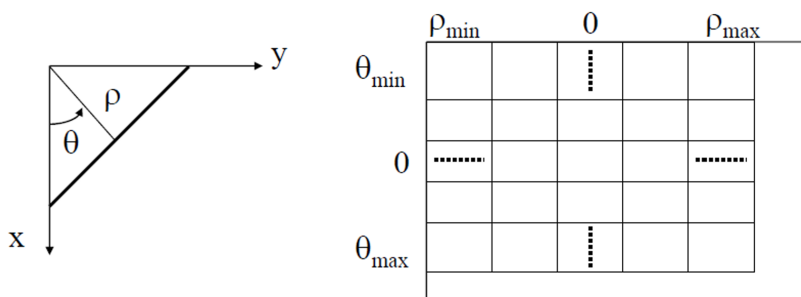
- The cell at co-ordinate (i,j) , with cell value $A(i,j)$, corresponds to the square associated with parameter space coordinates (a_i, b_j)
- The collection of accumulator cells is commonly called the Hough matrix (or Hough Array) and are computed as
 - Set all cells to zero
 - For every point (x_k, y_k) in the image plane, let a equal each of the allowed subdivision values on the a axis and solve for b using $b = -x_k a + y_k$
 - The resulting b 's are rounded off to the nearest allowed value in the b axis
 - If a choice of a_p results in solution b_q , we let $A(p,q) = A(p,q)+1$
 - At the end, a value of M in $A(i,j)$ corresponds to M points in the xy plane lying on the line $y = a_i x + b_j$

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Hough Transform – Accumulator Cells

- A problem with this representation is that the slope and intercept approach infinity as the line approaches the vertical
- Solution: Use the polar representation of a line given by

$$x \cos \theta + y \sin \theta = \rho$$



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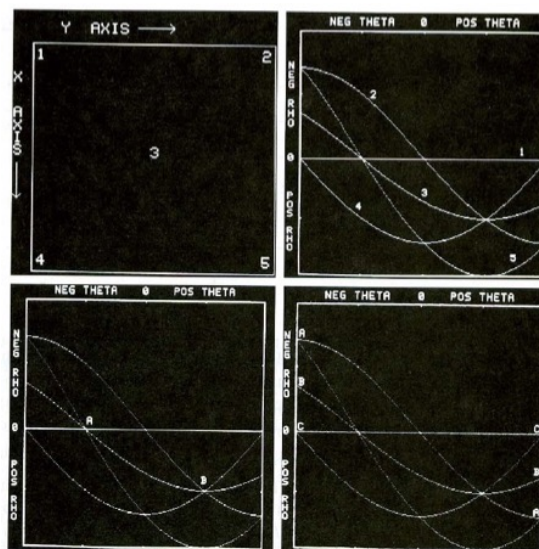
Hough Transform

- Instead of straight lines in the ab plane, we now have sinusoidal curves in the $\rho\theta$ plane
- M collinear points lying on the line

$$x \cos \theta_j + y \sin \theta_j = \rho_i$$
 yields M sinusoidal curves that intersect at (ρ_i, θ_j) in the parameter space
- The range of θ is $\pm 90^\circ$, measured with respect to the x -axis
 - A horizontal line has $\theta = 0^\circ$, with ρ equal to the positive x intercept
 - A vertical line has $\theta = +90^\circ$, with ρ equal to the positive y intercept or $\theta = -90^\circ$, with ρ equal to the negative y intercept
- The range of ρ is $\pm \frac{1}{2}D$. Where D is the distance between corners in the image.

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Hough Transform



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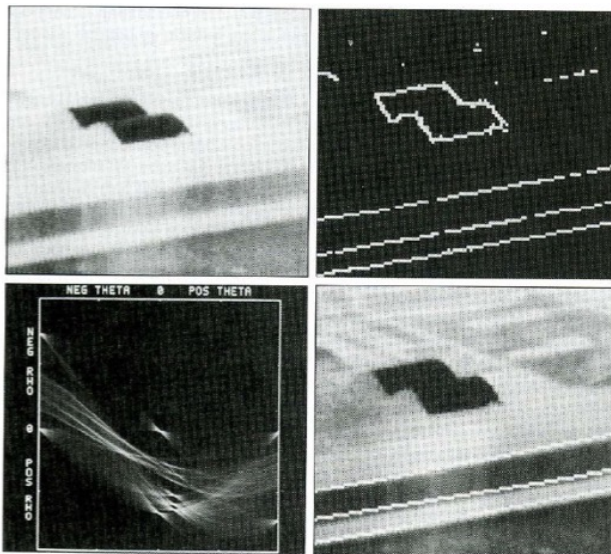
Hough Transform

Steps:

- Compute the gradient of an image and threshold it to obtain binary image.
- Specify subdivisions in the $\rho\theta$ plane
- Examine the counts of the accumulator cells for high pixel concentrations.
- Examine the relationship (principally for continuity) between pixels in a chosen cell.

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Hough Transform



- a. Infrared image
- b. Thresholded image
- c. Hough Transform
- d. Linked Pixel

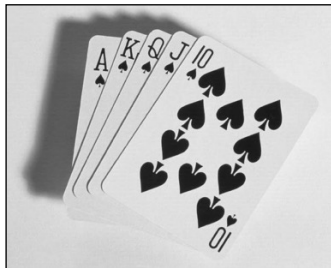
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Thresholding

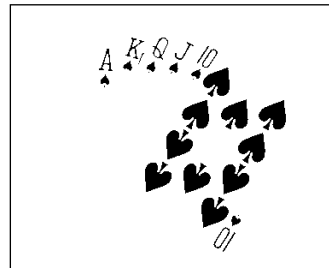
- Single value thresholding can be given mathematically as follows:

$$g(x, y) = \begin{cases} 1 & \text{if } f(x, y) > T \\ 0 & \text{if } f(x, y) \leq T \end{cases}$$

- For example: Imagine a poker playing robot that needs to visually interpret the cards in its hand



Original Image



Thresholded Image

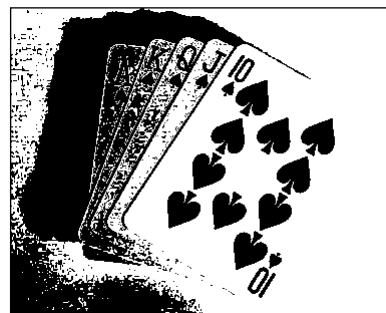
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Thresholding

If you get the threshold wrong the results can be disastrous



Threshold Too Low



Threshold Too High

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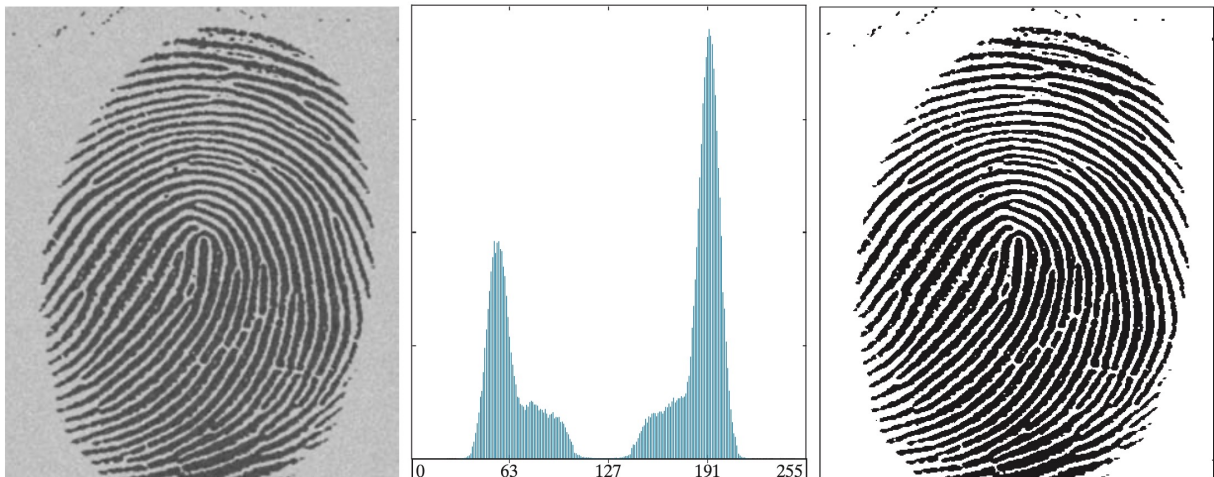
Global Thresholding

- Partition the image histogram using a single global threshold
- The success of this technique very strongly depends on how well the histogram can be partitioned
- The basic global threshold, T , is calculated as follows:
 - Select an initial estimate for T (the average grey level in the image)
 - Segment the image using T to produce two groups of pixels: G_1 consisting of pixels with grey levels $>T$ and G_2 consisting pixels with grey levels $\leq T$
 - Compute the average grey levels of pixels in G_1 to give μ_1 and G_2 to give μ_2
 - Compute a new threshold value:

$$T = \frac{\mu_1 + \mu_2}{2}$$
 - Repeat steps 2 – 4 until the difference in T in successive iterations is less than a predefined limit T_∞
- This algorithm works very well for finding thresholds when the histogram is suitable.

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Global Thresholding

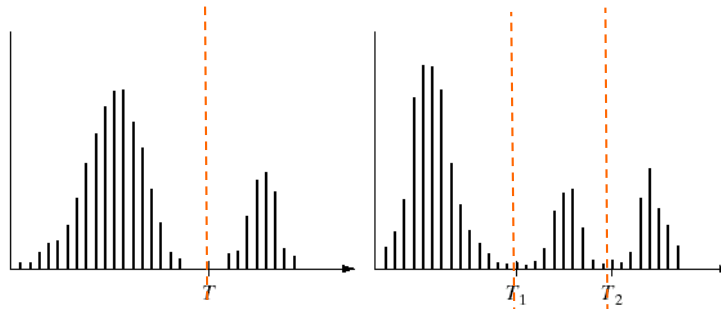


(a) Noisy fingerprint. (b) Histogram. (c) Segmented result using a global threshold

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Problem with Single Value Thresholding

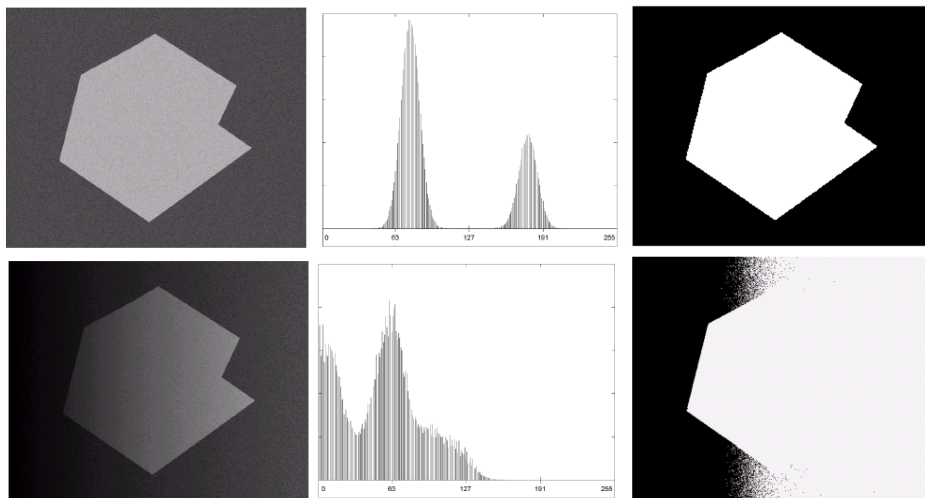
- Single value thresholding only works for bimodal histograms
- Suppose several objects with differing gray levels (with a dark background) comprise the image
- An object may be classified as belonging to one object class if $T_1 < f(x,y) \leq T_2$, to a second class if $f(x,y) > T_2$ or to background if $f(x,y) \leq T_1$



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Problem with Single Value Thresholding

Uneven illumination can really upset a single valued thresholding.



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Global, Local and Dynamic Thresholding

- Thresholding may be viewed as an operation that tests against a given function of the form

$$T = T(x, y, p(x, y), f(x, y))$$

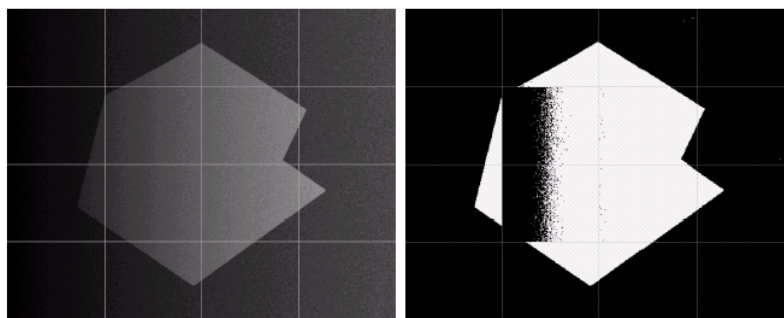
where $f(x, y)$ is the gray level of point (x, y) and $p(x, y)$ is some local property of the point - the average gray level of a neighborhood around (x, y)

- When T depends only on $f(x, y)$ the threshold is called **global** and if T depends on $f(x, y)$ and $p(x, y)$ the threshold is **local**
- If T depends on spatial coordinates (x, y) the threshold is called **dynamic**
- For example, a local threshold may be used if certain information about the nature of the objects in the image is known as priori and a dynamic threshold may be used in the case where object illumination is not uniform.

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Adaptive Thresholding

- An approach to handling situations in which single value thresholding will not work is to divide an image into sub images and threshold these individually
- Since the threshold for each pixel depends on its location within an image this technique is said to be adaptive



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Optimal Thresholding – Otsu's Algorithm

Otsu's algorithm may be summarized as follows:

- Compute the normalized histogram of the input image. Denote the components of the histogram by p_i , $i = 0, 1, 2, \dots, L - 1$.
- Compute the cumulative sums, $P_1(k)$, for $k = 0, 1, 2, \dots, L - 1$, using

$$P_1(k) = \sum_{i=0}^k P_i.$$
- Compute the cumulative means, $m(k)$, for $k = 0, 1, 2, \dots, L - 1$, using

$$m(k) = \sum_{i=0}^k i P_i$$
- Compute the global mean, m_G , using

$$m_G = \sum_{i=0}^{L-1} i P_i.$$
- Compute the between-class variance term, $\sigma_B^2(k)$, for $k = 0, 1, 2, \dots, L - 1$, using

$$\sigma_B^2(k) = \frac{[m_G P_1(k) - m(k)]^2}{P_1(k)[1 - P_1(k)]}$$

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Optimal Thresholding – Otsu's Algorithm

- Obtain the Otsu threshold, k^* , as the value of k for which $\sigma_B^2(k)$ is maximum. If the maximum is not unique, obtain k^* by averaging the values of k corresponding to the various maxima detected.
- Compute the global variance, σ_G^2 , using

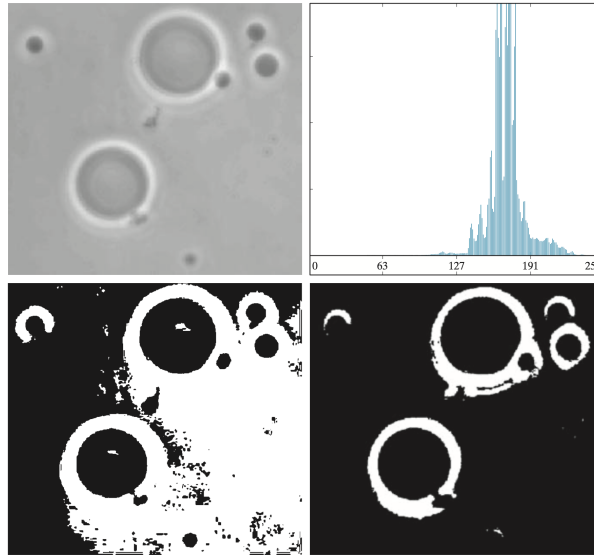
$$\sigma_G^2 = \sum_{i=0}^{L-1} (i - m_G)^2 p_i$$

and then obtain the separability measure, η^* , by evaluating given equation with $k = k^*$.

$$\eta(k) = \frac{\sigma_B^2(k)}{\sigma_G^2}$$

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Optimal Thresholding – Otsu's Algorithm



(a) Original image.
(b) Histogram
(high peaks
were clipped to
highlight details in
the lower values).
(c) Segmentation
result using
the basic global
thresholding
(d) Result using
Otsu's method.

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Using Edges to Improve Thresholding

- Important aspect of threshold selection: the ability to reliably identify mode peaks in a given histogram
- The chances of selecting a "good" threshold are enhanced if mode peaks are
 - tall
 - narrow
 - symmetric
 - and separated by deep valleys
- One approach for improving the histogram shape is to consider only those pixels that lie on or near a boundary between objects and the background

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Using Edges to Improve Thresholding

- An obvious advantage is that the histogram becomes less dependent on the size of objects in the image
- By choosing pixels on or near object boundaries (assuming an equal probability of choosing a pixel on the object or boundary) the histogram peaks tend to be made more symmetric
- Using pixels that satisfy some simple measures based on the gradient and Laplacian operators tends to deepen the valleys between histogram peaks

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Using Edges to Improve Thresholding

1. Determining if a pixel lies on a boundary: compute the gradient
 2. Determining what side, background (dark) or object (light), a pixel lies on: compute the Laplacian
- Using the gradient and Laplacian, a three-level image may be formed according to

$$s(x, y) = \begin{cases} 0 & \text{if } \nabla f < T \\ + \text{ if } \nabla f \geq T \text{ and } \nabla^2 f \geq 0 \\ - \text{ if } \nabla f \geq T \text{ and } \nabla^2 f < 0 \end{cases}$$

where 0, + and - are three distinct gray levels

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Using Edges to Improve Thresholding

- For a dark object on a light background, $s(x,y)$ is produced where
 - all pixels not on an edge are labeled 0
 - all pixels on the dark side of an edge are labeled +
 - all pixels on the light side of an edge are labeled -
- For a light object on a dark background, $s(x,y)$ is produced where
 - all pixels not on an edge are labeled 0
 - all pixels on the dark side of an edge are labeled -
 - all pixels on the light side of an edge are labeled +

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Using Edges to Improve Thresholding

- The information obtained with this procedure can be used to generate a segmented, binary image in which 1's correspond to objects of interest and 0's correspond to the background.
- The transition (along a horizontal or vertical scan line) from a light background to a dark object must be characterized by the occurrence of a - followed by a + in $s(x, y)$.
- The interior of the object is composed of pixels that are labeled either 0 or +. Finally, the transition from the object back to the background is characterized by the occurrence of a + followed by a -. Thus a horizontal or vertical scan line containing a section of an object has the following structure:

$$(\dots)(-, +)(0 \text{ or } +)(+, -)(\dots)$$

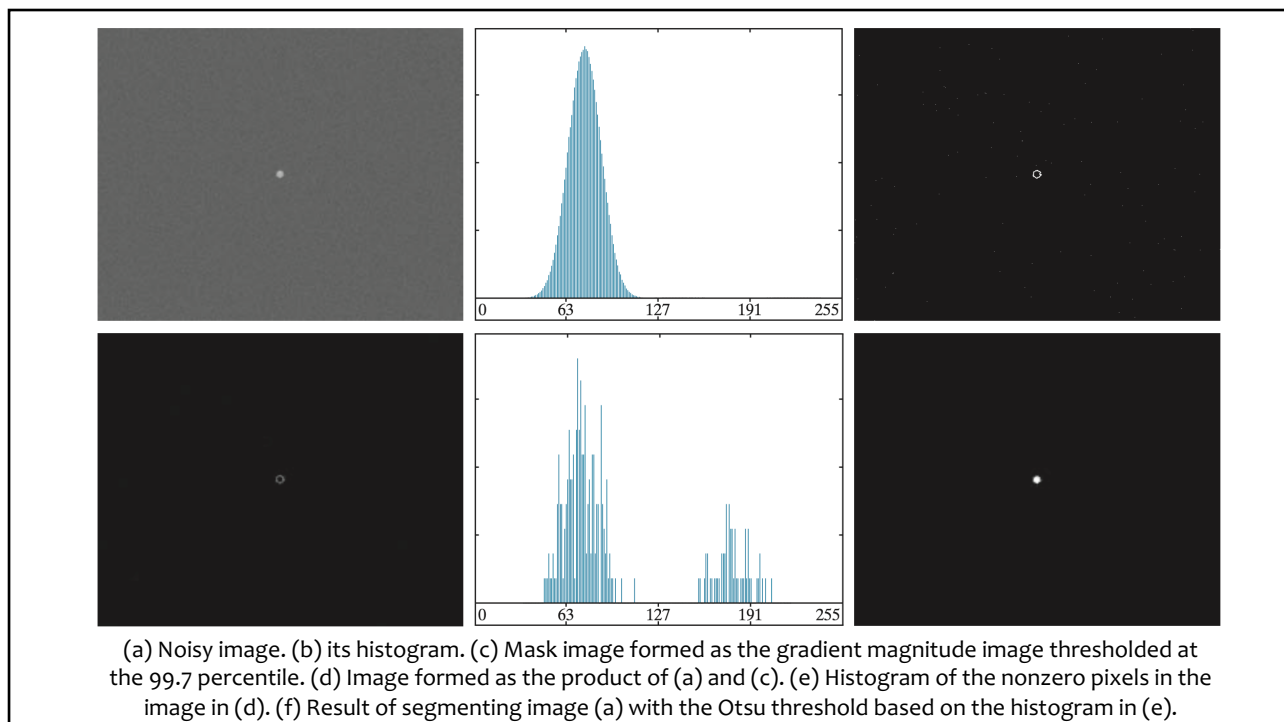
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Using Edges to Improve Thresholding

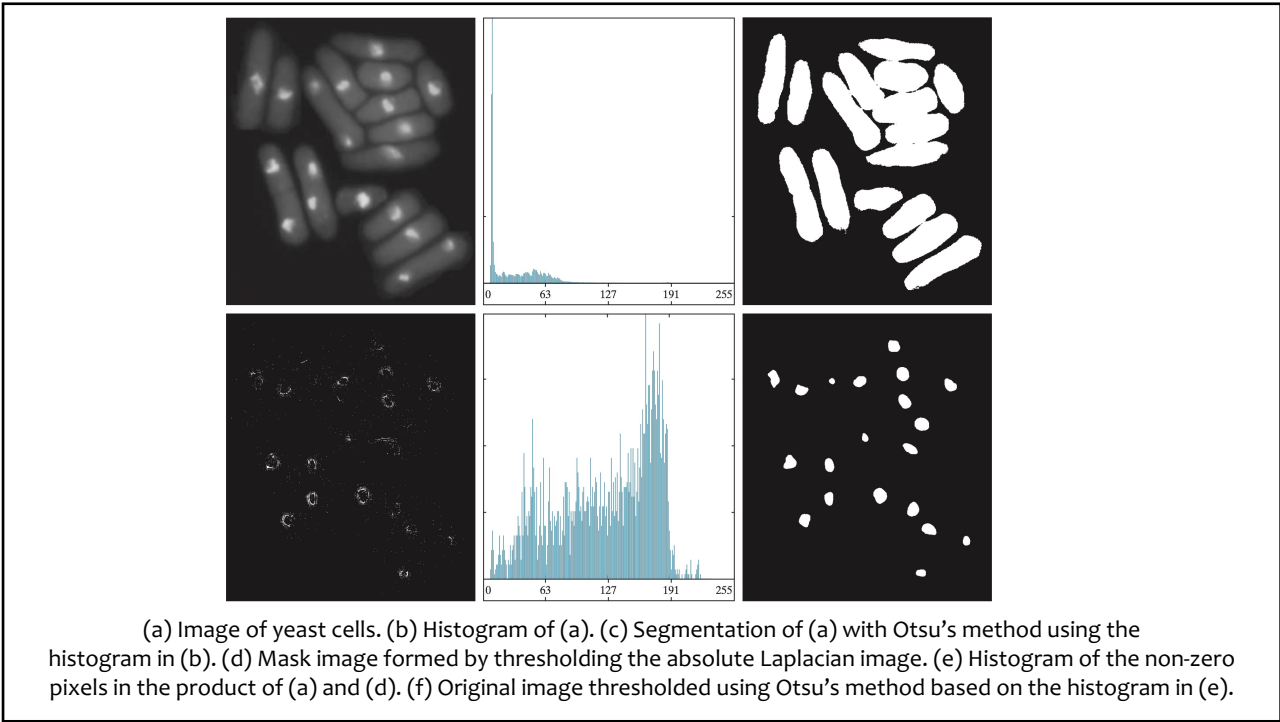
Summarized algorithm, where $f(x,y)$ is the input image:

1. Compute an edge image as either the magnitude of the gradient, or absolute value of the Laplacian, of $f(x,y)$.
2. Specify a threshold value, T .
3. Threshold the image from Step 1 using T from Step 2 to produce a binary image, $g_T(x,y)$. This image is used as a mask image in the following step to select pixels from $f(x,y)$ corresponding to “strong” edge pixels in the mask.
4. Compute a histogram using only the pixels in $f(x,y)$ that correspond to the locations of the 1-valued pixels in $g_T(x,y)$.
5. Use the histogram from Step 4 to segment $f(x,y)$ globally using, for example, Otsu’s method.

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Discussions and Classwork

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Lecture – 17

Harimohan Khatri

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In this lecture...

- i. Region Growing Based Segmentation
 - i. Seeded
 - ii. Unseeded
- ii. Region Splitting and Merging
- iii. Representation and Description
 - i. Chain Codes and Crack Codes
 - ii. Polygon Approximation
 - iii. Signature
 - iv. Shape Numbers
 - v. Fourier Descriptors

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Region Growing Based Segmentation

- Let:
 - $f(x,y) \rightarrow$ input image;
 - $S(x,y) \rightarrow$ seed array containing 1's at the locations of seed points and 0's elsewhere; and
 - $Q \rightarrow$ predicate to be applied at each location (x,y) . Arrays f and S are assumed to be of the same size.
- A basic region-growing algorithm based on 8-connectivity may be stated as follows.
 1. Find all connected components in $S(x,y)$ and reduce each connected component to one pixel; label all such pixels found as 1. All other pixels in S are labeled 0.

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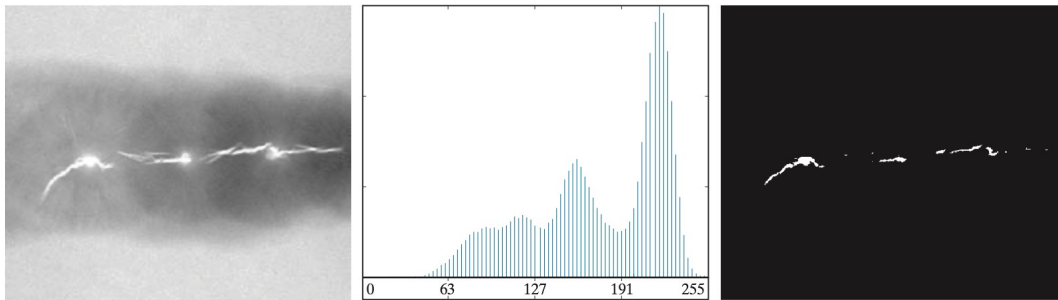
Region Growing Based Segmentation

2. Form an image f_Q such that, at each point (x,y) , $f_Q(x,y) = 1$ if the input image satisfies a given predicate, Q , at those coordinates, and $f_Q(x,y) = 0$ otherwise.

$$Q = \begin{cases} \text{TRUE} & \text{if the absolute difference of intensities} \\ & \text{between the seed and the pixel at } (x,y) \text{ is } \leq T \\ \text{FALSE} & \text{otherwise} \end{cases}$$
3. Let g be an image formed by appending to each seed point in S all the 1-valued points in f_Q that are 8-connected to that seed point.
4. Label each connected component in g with a different region label (e.g. integers or letters). This is the segmented image obtained by region growing.

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Region Growing Based Segmentation

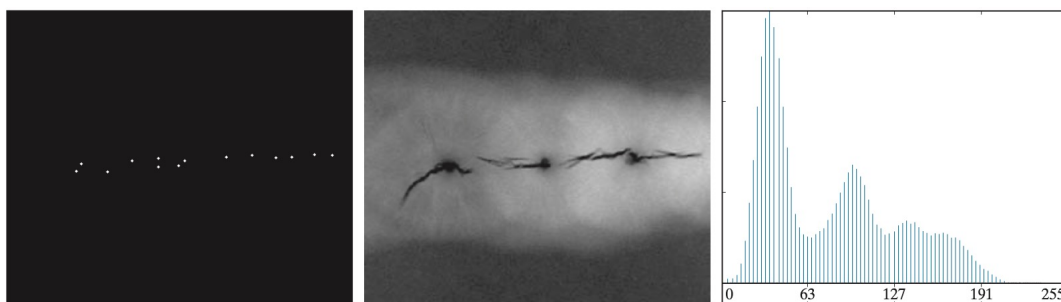


(a) X-ray image of a defective weld. (b) Histogram. (c) Initial seed image.

extract the seed points by thresholding the original image, using a threshold set at a high percentile (99%).

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Region Growing Based Segmentation



result of morphologically eroding each connected component in seeded image to a single point.

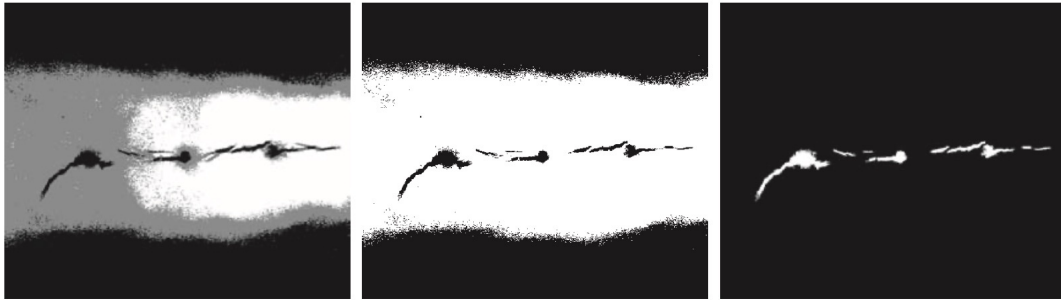
(d) Final seed image (the points were enlarged for clarity).

(e) Absolute value of the difference between the seed value (255) and (a).

(f) Histogram of (e).

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Region Growing Based Segmentation



- (g) Difference image thresholded using dual thresholds.
- (h) Difference image thresholded with the smallest of the dual thresholds.
- (i) Segmentation result obtained by region growing.

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Region Splitting and Merging

- Let R represent the entire image region and select a predicate Q .
- One approach for segmenting R is to subdivide it successively into smaller and smaller quadrant regions so that, for any region R_i , $Q(R_i) = \text{TRUE}$
- We start with the entire region, R . If $Q(R) = \text{FALSE}$, we divide the image into quadrants. If Q is FALSE for any quadrant, we subdivide that quadrant into sub-quadrants, and so on.
- This splitting technique has a convenient representation in the form of so-called quadtrees; images corresponding to the nodes of a quadtree are called quadregion or quadimages.

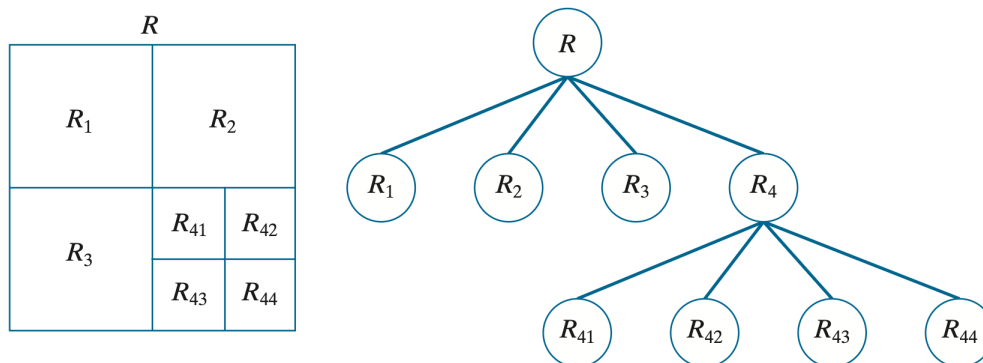
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Region Splitting and Merging

- After splitting final partition normally contains adjacent regions with identical properties.
- Merging adjacent regions whose combined pixels satisfy the predicate Q is required. That is, two adjacent regions R_i and R_k are merged only if $Q(R_i \cup R_k) = \text{TRUE}$.

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Region Splitting and Merging



- a) Partitioned image.
 b) Corresponding quadtree.
 R represents the entire image region.

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Region Splitting and Merging

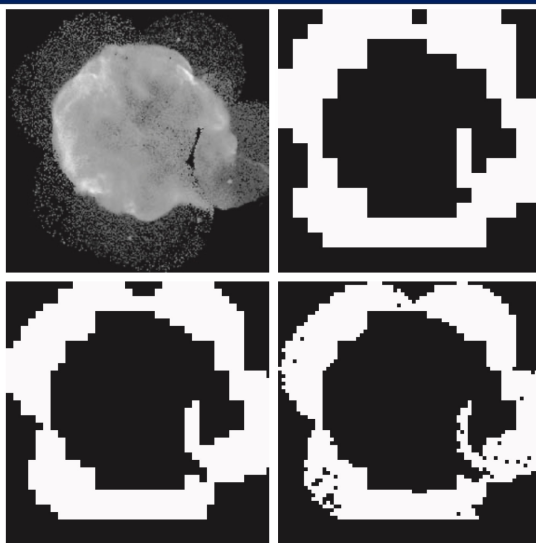
1. Split into four disjoint quadrants any region R_i for which $Q(R_i) = \text{FALSE}$.
2. When no further splitting is possible, merge any adjacent regions R_j and R_k for which $Q(R_j \cup R_k) = \text{TRUE}$.
3. Stop when no further merging is possible.

$$Q(R) = \begin{cases} \text{TRUE} & \text{if } \sigma_R > a \text{ AND } 0 < m_R < b \\ \text{FALSE} & \text{otherwise} \end{cases}$$

where σ_R and m_R are the standard deviation and mean of the region being processed, and a and b are nonnegative constants.

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Region Splitting and Merging



a) Image of the Cygnus Loop supernova, taken in the X-ray band by NASA's Hubble Telescope. (b) through (d) Results of limiting the smallest Allowed quad region to be of sizes of 32×32 , 16×16 , and 8×8 pixels, respectively.

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Representation and Description

- Result of segmentation need to be
 - Represented and
 - Described
- Representation: Transformation of raw image data into a format suitable for analysis.
- Description: Extracting quantitative features from representations that describe objects meaningfully.
- Together, they form the basis for object recognition and classification.
 - Reduces data complexity
 - Focuses on relevant object characteristics
 - Aids in feature-based analysis (e.g., classification, matching)

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Representation and Description

- Boundary Representation:
 - Focus on external shape or contour of object
 - Examples: Chain codes, Signatures
- Region-based Representation:
 - Focus on entire object region
 - Examples: Texture, statistical moments

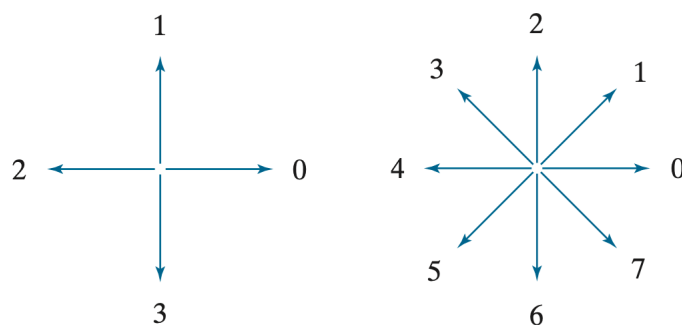
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Chain Codes

- A boundary to be encoded is overlaid on a square grid whose side length determines the resolution of the encoding.
- Starting at the beginning of the curve, the grid intersection points that come closest to it are used to define small line segments that join each grid point to one of its neighbors.
- The directions of these line segments are then encoded as small integers from zero to the number of neighbors used in encoding.

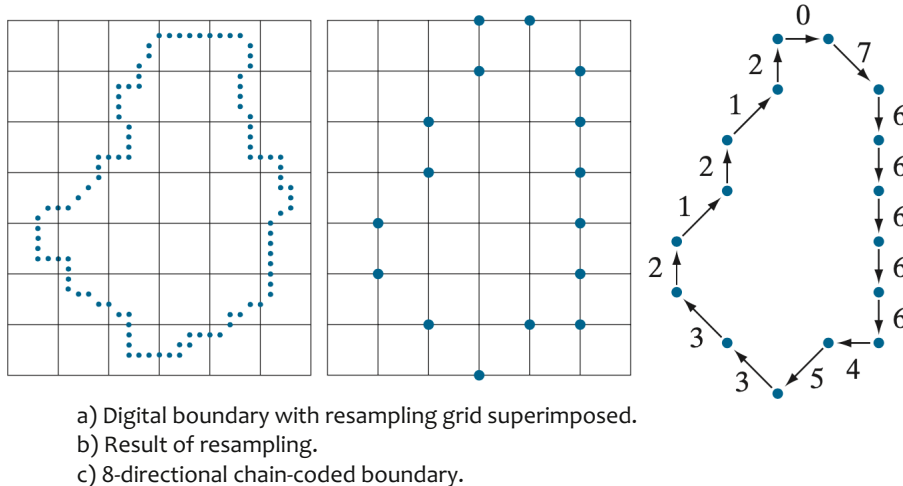
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Chain Codes



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Chain Codes



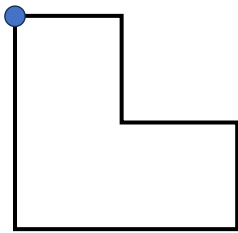
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Chain Codes

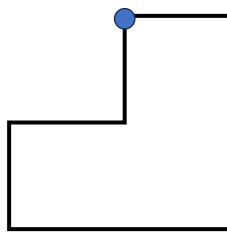
- The chain code of a boundary depends on the starting point.
- However, the code can be normalized by treating it as a circular sequence.
- Size (scale) normalization can be achieved by altering the size of the sampling grid.
- Rotation normalization can be achieved by using the first difference of the chain code instead of the code itself.
 - Count the number of direction changes (in counterclockwise direction) that separate two adjacent elements of the code

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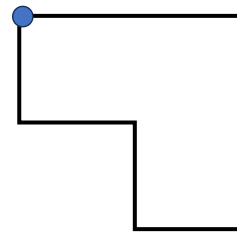
Chain Codes



CC → 03032211
 FCC → 31330303
 NFCC → 03033133



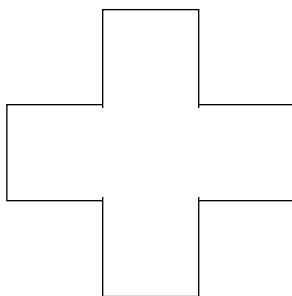
CC → 03322101
 FCC → 30303313
 NFCC → 03033133



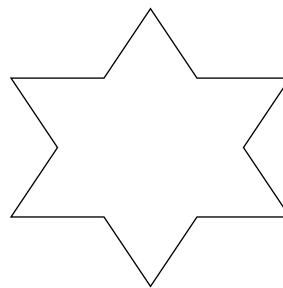
CC → 00332121
 FCC → 03033133
 NFCC → 03033133

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Chain Codes – Exercise



Apply 4 – Directional Chain Code

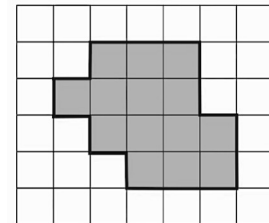
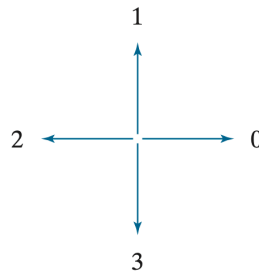


Apply 8 – Directional Chain Code

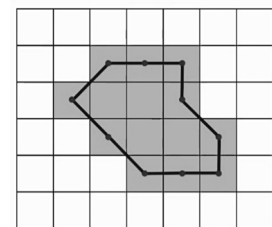
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Crack Codes

- Represents the boundary between two adjacent pixels of different values (typically object vs background).
- Based on the edges between pixels, not the pixel centers.
- Path follows the cracks between object and background
- Only 4- Directional code



Crack Code



8 – Directional Chain Code

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Polygon Approximation

- A digital boundary can be approximated with arbitrary accuracy by a polygon.
- For a closed curve, the approximation becomes exact when the number of segments of the polygon is equal to the number of points in the boundary, so each pair of adjacent points defines a segment of the polygon.
- The goal of a polygonal approximation is to capture the essence of the shape in a given boundary using the fewest possible number of segments.
- One of the algorithm used for boundary approximation is *minimum-perimeter polygon (MPP)*

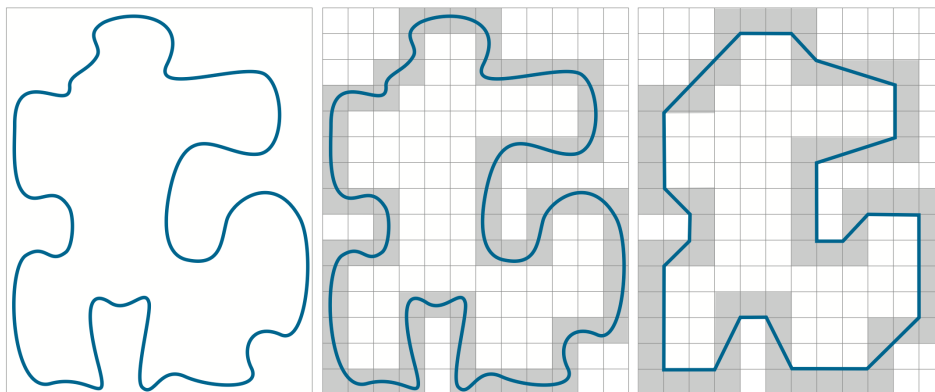
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Polygon Approximation

- An intuitive approach for computing MPPs is to enclose a boundary by a set of concatenated cells.
- Think of the boundary as a rubber band contained in the gray cells. As it is allowed to shrink, the rubber band will be constrained by the vertices of the inner and outer walls of the region of the gray cells.
- Ultimately, this shrinking produces the shape of a polygon of minimum perimeter (with respect to this geometrical arrangement) that circumscribes the region enclosed by the cell strip.

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Polygon Approximation



- (a) An object boundary.
(b) Boundary enclosed by cells (shaded).
(c) Minimum-perimeter polygon obtained by allowing the boundary to shrink. The vertices of the polygon are created by the corners of the inner and outer walls of the gray region.

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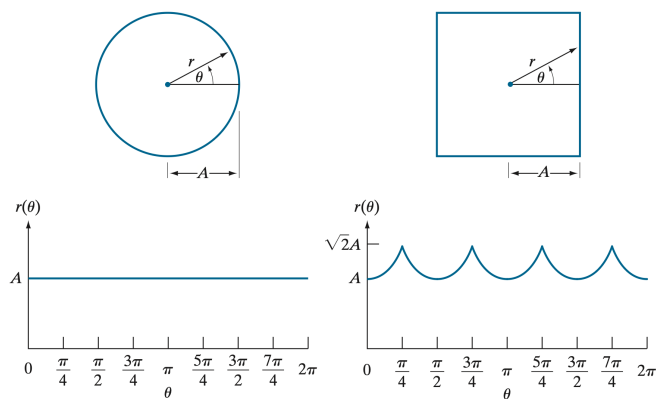
Signatures

- A signature is a 1-D functional representation of a 2-D boundary
- Plot the distance from the centroid to the boundary as a function of angle
- The basic idea of using signatures is to reduce the boundary representation to a 1-D function that presumably is easier to describe than the original 2-D boundary.
- Based on the assumptions of uniformity in scaling with respect to both axes, and that sampling is taken at equal intervals of θ , changes in the size of a shape result in changes in the amplitude values of the corresponding signature.

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Signatures

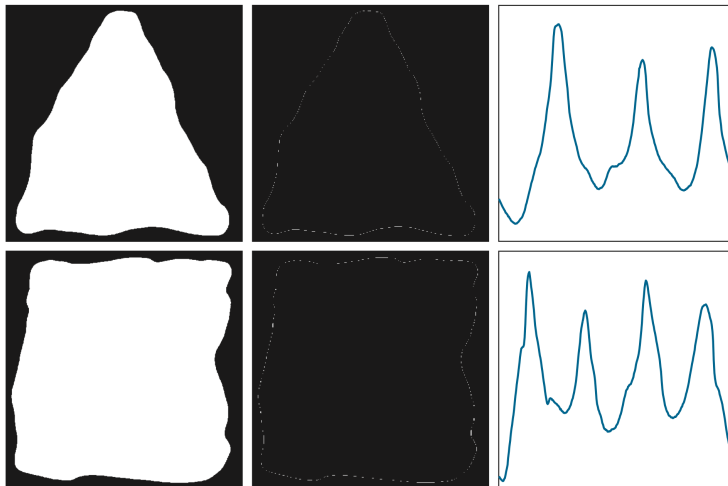
- The main advantage of this method is simplicity, but it has the disadvantage that scaling of the entire function depends on only two values: the minimum and maximum.
- If the shapes are noisy, this can be a source of significant error from object to object.



Distance-versus-angle signatures.

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Signatures



- Two binary regions,
- Their external boundaries, and
- Their corresponding signatures.

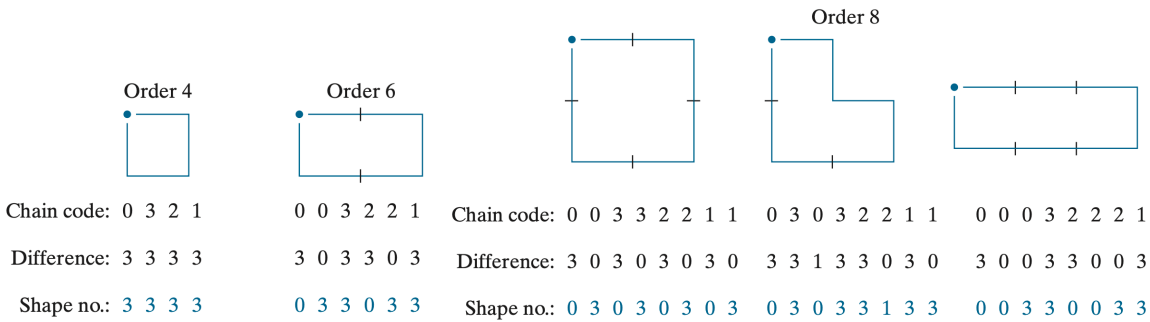
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Shape Numbers

- The shape number of a Freeman chain-coded boundary, based on the 4-directional code is defined as the first difference of smallest magnitude.
- The order, n , of a shape number is defined as the number of digits in its representation.
 - n is even for a closed boundary, and its value limits the number of possible different shapes.

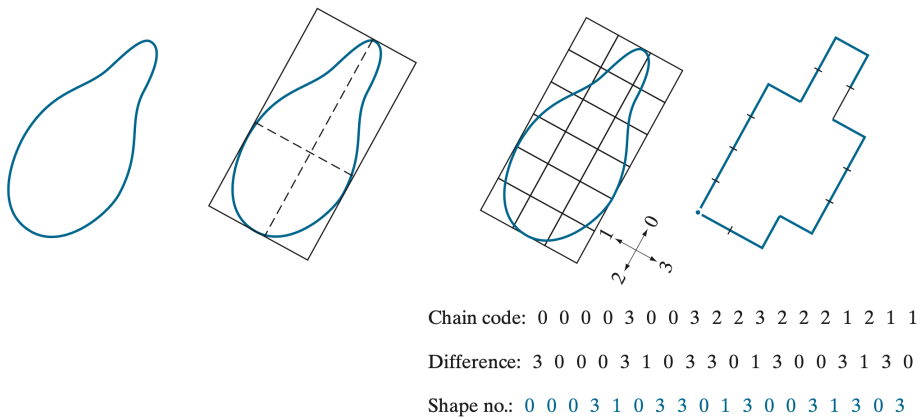
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Shape Numbers



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Shape Numbers



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Fourier Descriptors

- Let a shape on xy-plane has k points on its boundary.
- Starting at an arbitrary point (x_0, y_0) coordinate pairs $(x_0, y_0), (x_1, y_1), (x_2, y_2), (x_3, y_3), \dots, (x_{K-1}, y_{K-1})$ are encountered in traversing the boundary, say, in the counterclockwise direction.
- These coordinates can be expressed in the form $x(k) = x_k$ and $y(k) = y_k$
- Using this notation, the boundary itself can be represented as the sequence of coordinates $s(k) = [x(k), y(k)]$ for $k = 0, 1, 2, 3, \dots, K-1$
- And each coordinate pair can be treated as a complex number so that

$$s(k) = x(k) + jy(k) \quad \text{for } k = 0, 1, 2, \dots, K-1$$

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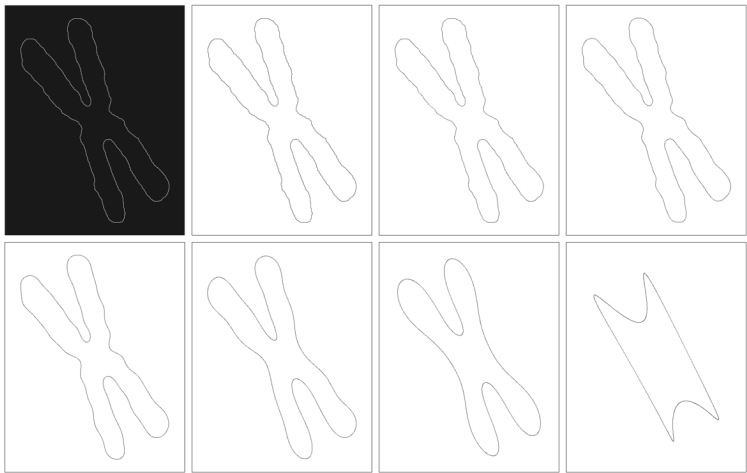
Fourier Descriptors

- By computing the DFT of complex sequence we get the Fourier Descriptors as:

$$a(u) = \sum_{k=0}^{K-1} s(k) e^{-j2\pi uk/K} \quad \text{for } u = 0, 1, 2, \dots, K-1$$

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Fourier Descriptors



(a) Boundary of a human chromosome (2868 points).
(b)–(h) Boundaries reconstructed using 1434, 286, 144, 72, 36, 18, and 8 Fourier descriptors, respectively.

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Discussions and Classwork

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